Year 12 Maths - Integration

Similar questions - Definite integrals and checking methods

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1

(a) Show that $\int_1^a \left(2x - \frac{1}{x^2}\right) \mathrm{d}x = a^2 + \frac{1}{a} - 2$, where $a > 1$. **[3]**

(b) Given that this integral equals $\frac{5}{2}$, show that $2a^3 - 9a + 2 = 0$. **[2]**

(c) Show that $a = 2$ is a root of this cubic, and show that the cubic has no other root greater than 1. **[5]**

2 Priya and Leo integrate $(2x-1)^2$ with respect to x.

Priya: $\int (2x-1)^2 \,\mathrm{d}x = \frac{(2x-1)^3}{6} + c$ Leo: $\int (4x^2 - 4x + 1) \,\mathrm{d}x = \frac{4x^3}{3} - 2x^2 + x + c$

(a) Show, by differentiating, that Priya's answer is correct. **[2]**

(b) Explain why the two answers look different even though both are correct. **[3]**

(c) Evaluate $\int_0^1 (2x-1)^2 \,\mathrm{d}x$ using each student's method and show that they agree. **[3]**

Question 1

(a) $\left[x^2+\frac{1}{x}\right]_1^a = \left(a^2+\frac{1}{a}\right)-(1+1) = a^2+\frac{1}{a}-2$

(b) $a^2+\frac{1}{a}-2=\frac{5}{2}$, so $a^2+\frac{1}{a}=\frac{9}{2}$. Multiply by $2a$: $2a^3+2=9a$, so $2a^3-9a+2=0$

(c) $2(8)-18+2=0$. Factorise: $2a^3-9a+2=(a-2)(2a^2+4a-1)$. The quadratic has roots $a=\frac{-2\pm\sqrt{6}}{2}$, approximately 0.22 and -2.22, both less than 1. So $a=2$ is the only root with $a>1$

Question 2

(a) $\frac{\mathrm{d}}{\mathrm{d}x}\left[\frac{(2x-1)^3}{6}\right]=\frac{3(2x-1)^2\times 2}{6}=(2x-1)^2$

(b) Expanding Priya's: $\frac{8x^3-12x^2+6x-1}{6}=\frac{4x^3}{3}-2x^2+x-\frac{1}{6}$. The two answers differ only by a constant, which is absorbed into c

(c) Priya: $\frac{1^3-(-1)^3}{6}=\frac{1}{3}$. Leo: $\left(\frac{4}{3}-2+1\right)-0=\frac{1}{3}$. They agree (the constants cancel in a definite integral)